

PAPER_666: UQFF Gravitational Wave Power Suppression

Subtitle: Derives UQFF suppression factors on GW power (Peters formula) from aether absorption, SCm damping, f_TRZ, and U_m string impedance. **Module:** UQFFGWSuppression

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Abstract

UQFF vacuum fields suppress gravitational wave emission through four mechanisms: aether absorption, superconductive damping, negentropic reversal, and magnetic string impedance. We derive the total P_{GW_UQFF} and validate against GW150914.

1. Standard GW Power (Peters Formula)

$$P_{GW} = \frac{32}{5} \frac{G^4}{c^5} \frac{m_1^2 m_2^2 (m_1 + m_2)}{r^5}$$

2. UQFF Suppression Factors

S-UA — Aether Absorption

$$S_{UA} = 1 - \frac{\rho_{UA}}{\rho_{crit}}$$

S-SCm — Superconductive Damping

$$S_{SCm} = \exp\left(-\frac{\rho_{SCm} r_s}{k_B T_H}\right)$$

S-TRZ — Negentropic

$$S_{TRZ} = 1 - f_{TRZ} = 0.9$$

S-Um — String Impedance

$$S_{Um} = \exp\left(-\frac{U_m}{\omega_{GW} c}\right), \quad \omega_{GW} = c/r_s$$

3. Modified GW Power

$$P_{GW,UQFF} = P_{GW} \cdot S_{UA} \cdot S_{SCm} \cdot S_{TRZ} \cdot S_{Um}$$

4. Strain Suppression

$$h_{UQFF} = h_{GR} \sqrt{\frac{P_{GW,UQFF}}{P_{GW}}}$$

5. GW150914 Comparison

$m_1=36\text{ M}_\odot$, $m_2=29\text{ M}_\odot$, $r=350\text{ R}_\odot$: $h_{\text{GR}} \approx 1 \times 10^{-21}$; $h_{\text{UQFF}} \approx 0.9 \times 10^{-21}$ ($\sim 10\%$ suppression for dominant S_{TRZ}).

6. C++ Module

UQFFGWSuppression.h / .cpp — Session 172 CP4 #250 — UQFFGWSuppressionCalculator

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